

RATIO ANALYSIS

1. Meaning of Ratio

It is a mathematical expression of the relationship between two accounting figures.

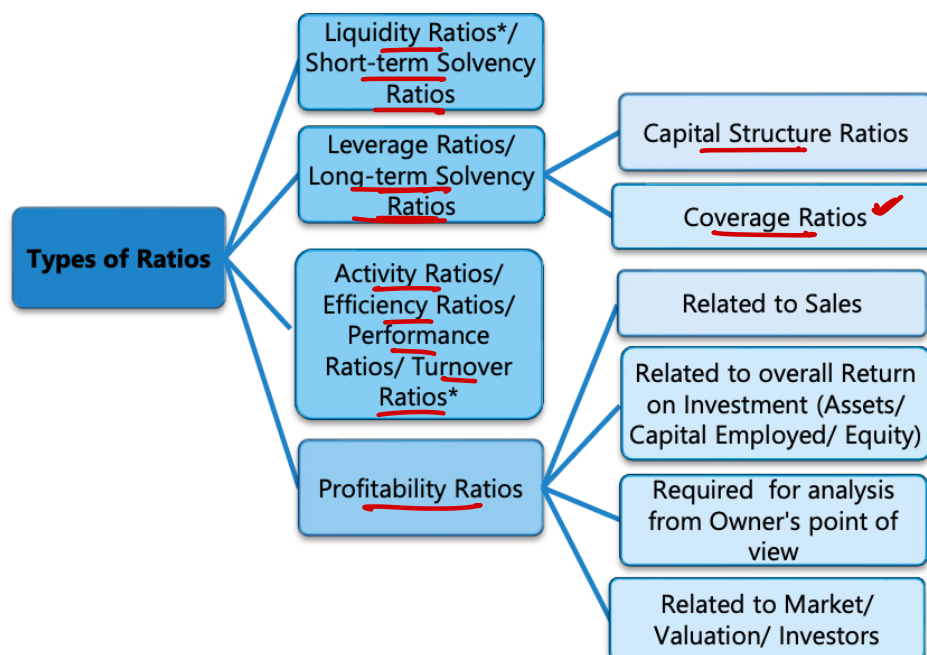
2. Ratio Analysis

Ratio analysis is the process of identifying the financial strengths and weaknesses of the enterprise by logically establishing relationship between the items of Balance Sheet or Income Statement or both and interpreting the results thereof in order to derive meaningful conclusions

3. Type of Ratios

The ratios may be classified on the basis of requirements of various users. The classification is as follows:

- (a) Liquidity Ratios or Short-term Solvency Ratios
- (b) Long-term Solvency Ratios or Leverage Ratios
- (c) Activity Ratios or Efficiency Ratios or Turnover Ratios or Performance Ratios
- (d) Profitability Ratios



4. Liquidity Ratios or Short-term Solvency Ratios

These ratios are used to evaluate the firm's short term solvency.

$$(a) \text{ Current Ratio } = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$(b) \text{ Quick Ratio } = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick Assets = Current Assets – Inventory – Prepaid Expenses

$$(c) \text{ Absolute Cash Ratio } = \frac{\text{Cash \& Bank Balance} + \text{Marketable Securities}}{\text{Current Liabilities}}$$

$$= \frac{\text{Cash \& Bank Balance} + \text{Current Invest.}}{\text{Current Liabilities}}$$

$$(d) \text{ Basic Defense Interval/Interval Measure }$$

$$= \frac{\text{Quick Assets}}{\text{Daily Cash Operating Expenses}}$$

$$= \frac{\text{Cash \& Bank Balance} + \text{Marketable Securities} + \text{Net Receivables}}{\text{Daily Cash Operating Expenses}}$$

$$\text{Daily Cash Operating Expenses} = \frac{\text{COGS} + \text{Admin.} + \text{S\&D \& General Exp.} - \text{Non cash Expenses}}{\text{No. of days in a year}}$$

$$(e) \text{ Net Working Capital } = \text{Current Assets} - \text{Current Liabilities (excluding short-term bank borrowing)}$$

5. Long Term Solvency Ratios or Leverage Ratios

These ratios are used to evaluate firm's long term solvency.

(I) Capital Structure Ratios

$$(a) \text{ Equity Ratio } = \frac{\text{Shareholder's Equity}}{\text{Net Assets/Capital Employed}}$$

$$(b) \text{ Debt Ratio } = \frac{\text{Total Debt}}{\text{Net Assets/Capital Employed}}$$

$$(c) \text{ Debt – Equity Ratio } = \frac{\text{Long-term Debts}}{\text{Shareholder's Equity}} = \frac{\text{Total Debts}}{\text{Shareholder's Equity}}$$

$$(d) \text{ Debt to Assets Ratio } = \frac{\text{Total Debt}}{\text{Total Assets}}$$

$$(e) \text{ Capital Gearing Ratio } = \frac{\text{Funds Bearing Fixed Financial Payments}}{\text{Equity Shareholder's Funds}}$$

$$= \frac{\text{Preference Share Capital} + \text{Debentures} + \text{Other borrowed funds}}{\text{Equity Shareholder's Funds}}$$

$$(f) \text{ Proprietary Ratio } = \frac{\text{Proprietor's Fund}}{\text{Total Assets}} \times 100$$

(II) Coverage Ratios

$$(a) \text{ Interest Coverage Ratio} = \frac{EBIT}{Interest}$$

$$(b) \text{ Preference Dividend Coverage Ratio} = \frac{EAT}{Preference Dividend}$$

$$(c) \text{ Equity Dividend Coverage Ratio} = \frac{EAE}{Equity Dividend}$$

$$(d) \text{ Debt Service Coverage Ratio} = \frac{\text{Earnings available for debt service}}{\text{Interest} + \text{Principal portion of installment}}$$

$$(e) \text{ Fixed Charges Coverage Ratio} = \frac{EBIT + \text{Non-cash Operating Expenses}}{\text{Interest} + \text{Principal portion of installment}}$$

(PAT + Non-cash + Int. Exp.)

6. Points to Remember (PTRs)➤ **Equity Shareholder Fund / Shareholder's Equity**

= Equity share capital + Reserve & Surplus – fictitious assets

➤ **Shareholder's Fund/ Owners Fund/ Proprietary Fund/ Net Worth**

= Equity Shareholder Fund + Preference Share Capital

➤ **Capital Employed** = Shareholder fund + Long term Debt – Non Trade Investments
- Non Trade Loans & Advance – Capital WIP➤ **Capital Employed** = Fixed Assets + Working Capital + Long term trade investment➤ **Total Debt or Total Outside Liabilities**

= Long term borrowing + Short term borrowing (Current liabilities)

➤ Total Assets must be excluding fictitious assets

7. Activity or Efficiency or Performance or Turnover or Velocity Ratio

These ratios help in commenting on the efficiency of the enterprise in managing its assets.

- (a) **Total Assets Turnover Ratio** = $\frac{\text{Sales/COGS}}{\text{Total Assets}}$
- (b) **Fixed Assets Turnover Ratio** = $\frac{\text{Sales/COGS}}{\text{Fixed Assets}}$
- (c) **Capital/Net Assets Turnover Ratio** = $\frac{\text{Sales/COGS}}{\text{Capital Employed}}$
- (d) **Current Assets Turnover Ratio** = $\frac{\text{Net Sales/COGS}}{\text{Current Assets}}$
- (e) **Working Capital Turnover Ratio** = $\frac{\text{Net Sales/COGS}}{\text{Working Capital}}$
- (f) **Inventory (Finsihed Stock) Turnover Ratio** = $\frac{\text{Cost of Goods Sold/Sales}}{\text{Average FG Stock}}$
- (g) **Inventory (Finished Stock)Velocity** = $\frac{\text{Average FG Stock}}{\text{Average Cost of Goods Sold per day}}$

$$= \frac{12 \text{ months} / 52 \text{ weeks} / 365 \text{ days}}{\text{Stock Turnover Ratio}}$$
- (h) **Inventory (Raw Material) Turnover Ratio** = $\frac{\text{Raw Material Consumed}}{\text{Average RM Stock}}$
- (i) **Inventory (Finished Stock)Velocity** = $\frac{\text{Average RM Stock}}{\text{Average RM Consumed per day}}$

$$= \frac{12 \text{ months} / 52 \text{ weeks} / 365 \text{ days}}{\text{Inventory Turnover Ratio}}$$
- (j) **Receivables Turnover Ratio** = $\frac{\text{Credit Sales}}{\text{Average account receivables}}$
- (k) **Receivables Velocity** = $\frac{\text{Average Receivables}}{\text{Average Credit Sales per day}}$

$$= \frac{12 \text{ months} / 52 \text{ weeks} / 365 \text{ days}}{\text{Receivables Turnover Ratio}}$$
- (l) **Payables Turnover Ratio** = $\frac{\text{Credit Purchases}}{\text{Average account payables}}$
- (m) **Payables Velocity** = $\frac{\text{Average Payables}}{\text{Average Credit Purchases per day}}$

$$= \frac{12 \text{ months} / 52 \text{ weeks} / 365 \text{ days}}{\text{Payables Turnover Ratio}}$$

8. Points to Remember (PTRs)

- Sales will be excluding GST and net of returns if any.
- It is prefer to take average figure in denominator. If average can't be computed then consider closing values.

9. Profitability Ratios

These are used to measure the firm's operating efficiency or profitability.

These are further classified in four categories:

- (A) Related to Sales
- (B) Related to Overall Return on Assets or Investment
- (C) From Owner's Point of View
- (D) Related to Market or Valuation or Investors

10. Profitability Ratios Related to Sales

- (a) **Gross Profit Ratio** = $\frac{\text{Gross Profit}}{\text{Net Sales}} \times 100$
- (b) **Net Profit Ratio (after tax)** = $\frac{\text{Net profit or EAT}}{\text{Sales}} \times 100$
- (c) **Net Profit Ratio (before tax)** = $\frac{\text{EBT}}{\text{Sales}} \times 100$
- (d) **Operating Profit Ratio** = $\frac{\text{Operating Profit or EBIT}}{\text{Sales}} \times 100$
- (e) **Cost of Goods Sold Ratio** = $\frac{\text{Cost of Goods Sold}}{\text{Net Sales}} \times 100$
- (f) **Operating Expense Ratio** = $\frac{\text{Operating Expenses}}{\text{Sales}} \times 100$
- (g) **Operating Ratio** = $\frac{\text{COGS} + \text{Operating Expenses}}{\text{Sales}} \times 100$
- (h) **Financial Expenses Ratio** = $\frac{\text{Financial Expenses}}{\text{Sales}} \times 100$

11. Profitability Ratios related to Overall Return on Investment or Assets

- (a) **Return on Assets (ROA)** =
$$\frac{\text{Net Profit after Tax}}{\text{Average Total/Tangible/Fixed Assets}}$$

$$= \frac{\text{EBIT}(1-t)}{\text{Average Total/Tangible/Fixed Assets}}$$

$$= \frac{\text{PAT} + \text{Interest}}{\text{Average Total/Tangible/Fixed Assets}}$$
- (b) **Return on Capital Employed (ROCE) (Before Tax)** =
$$\frac{\text{EBIT}}{\text{Average Capital Employed}} \times 100$$
- (c) **Return on Capital Employed (ROCE) (After Tax)** =
$$\frac{\text{EBIT}(1-t)}{\text{Average Capital Employed}} \times 100$$
- (d) **Return on Equity (ROE)** =
$$\frac{\text{PAT} - \text{Preference Dividend}}{\text{Equity Shareholder's Fund}} \times 100$$

12. Profitability Ratios from Owner's Point of View

- (a) **Earning per Share (EPS)** =
$$\frac{\text{Earning for Equity}}{\text{No. of Equity Shares}}$$
- (b) **Dividend Per Share (DPS)** =
$$\frac{\text{Equity Dividend}}{\text{No. of Equity Shares}}$$
- (c) **Dividend Payout Ratio (DP Ratio)** =
$$\frac{\text{DPS}}{\text{EPS}} \times 100$$

13. Profitability Ratios related to Market or Valuation or Investors

- (a) **Price Earnings Ratio (PE Ratio)** =
$$\frac{\text{Market Price per share (MPS)}}{\text{Earning Per Share (EPS)}}$$
- (b) **Dividend Yield Ratio** =
$$\frac{\text{DPS}}{\text{MPS}} \times 100$$
- (c) **Earning Yield Ratio** =
$$\frac{\text{EPS}}{\text{MPS}} \times 100$$
- (d) **Market Value / Book Value Per share (MV / BV)** =
$$\frac{\text{Market Value per share}}{\text{Book Value per share}}$$
- (e) **Q Ratio** =
$$\frac{\text{Market value of equity and liabilities}}{\text{Estimated replacement cost of assets}}$$

14. Du Pont Model

- (A) **Return on Capital Employed (ROCE)** = Operating Profit Margin × Capital Turnover
- (B) **Return on Equity (ROE)** = Net Profit Margin × Assets Turnover × Equity Multiplier

Question – 1

X Co. has made plans for the next year. It is estimated that the company will employ total assets of ₹ 8,00,000; 50 per cent of the assets being financed by borrowed capital at an interest cost of 8 percent per year. The direct costs for the year are estimated at ₹ 4,80,000 and all other operating expenses are estimated at ₹ 80,000. The goods will be sold to customers at 150 per cent of the direct costs. Tax rate is assumed to be 50 per cent.

You are required to calculate:

- (i) Operating profit margin (before tax) $\rightarrow \frac{1.608}{7.208} = 22.22\%$
- (ii) Net profit margin (after tax) $\rightarrow \frac{64000}{7.208} = 8.89\%$
- (iii) Return on assets (on operating profit after tax) $\rightarrow \frac{1.608(1-0.50)}{8.2} = 10\%$
- (iv) Assets turnover $\rightarrow 7.208 \div 82 = 0.9$
- (v) Return on owner's equity $\rightarrow \frac{\text{EAT}}{\text{Sh. fund}} = \frac{0.648}{(82 \times 50\%)} = 16\%$

Solution

The net profit is calculated as follows:

Particulars	₹
Sales (150% × 4,80,000)	7,20,000
Direct costs	(4,80,000)
Gross Profit	2,40,000
Operating expenses	(80,000)
Profit before interest and tax (EBIT)	1,60,000
Interest charges (8% of 4,00,000)	(32,000)
Profit before tax	1,28,000
Taxes @ 50%	(64,000)
Net profit after taxes	64,000

- (i) Operating profit margin = $\frac{\text{EBIT}}{\text{Sales}} = \frac{₹1,60,000}{₹7,20,000} = 0.2222$ or 22.22%
- (ii) Net profit margin = $\frac{\text{Net Profit after taxes}}{\text{Sales}} = \frac{₹64,000}{₹7,20,000} = 0.089$ or 8.9%
- (iii) Return on assets = $\frac{\text{EBIT}(1-T)}{\text{Assets}} = \frac{₹1,60,000(1-0.5)}{8,00,000} = 0.10$ or 10%
- (iv) Asset turnover = $\frac{\text{Sales}}{\text{Assets}} = \frac{₹7,20,000}{₹8,00,000} = 0.9$ times
- (v) Return on equity = $\frac{\text{Net Profit after taxes}}{\text{Owners' equity}} = \frac{₹64,000}{50\% \text{ of } ₹8,00,000} = \frac{₹64,000}{₹4,00,000} = 16\%$

Question – 2

CA = 200 CL = 100

Assuming the current ratio of a company is 2, STATE in each of the following cases whether the ratio will improve or decline or will have no change:

- (a) Payment of current liability $\rightarrow \frac{200-10}{100-10} = \frac{190}{90} = 2.11 \rightarrow \text{Improve}$
- (b) Purchase of fixed assets by cash $\rightarrow \frac{200-10}{100} = \frac{190}{100} = 1.9 \rightarrow \text{Decline}$
- (c) Cash collected from customers $\rightarrow \frac{200+10-10}{100} = 2 \rightarrow \text{No Ch.}$
- (d) Bill receivable dishonored $\rightarrow \frac{200+10-10}{100} = 2 \rightarrow \text{No Ch.}$
- (e) Issue of new shares $\rightarrow \frac{200+10}{100} = \frac{210}{100} = 2.1 \rightarrow \text{Improve}$

Solution

Given, Current ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2$

S. No.	Situation	Improve/Decline/No Change	Reason
(a)	Payment of current liability	Current ratio will improve	Let us assume CA is ₹ 2 lakhs & CL is ₹ 1 lakhs. If payment of CL is ₹ 10,000 then CA = 1,90,000 and CL = 90,000 Current ratio = $\frac{1,90,000}{90,000} = 2.11$ Payment of current liability will reduce the same amount in the numerator and denominator. Hence, the ratio will improve.
(b)	Purchase of fixed assets by cash	Current ratio will decline	Since the cash being a current asset converted into fixed asset, current assets reduced, thus current ratio will fall.
(c)	Cash collected from customers	Current ratio will not change	Cash will increase and debtors will reduce. Hence no change in current assets.
(d)	Bills receivable dishonored	Current ratio will not change	Bills receivable will come down and debtors will increase. Hence no change in current assets.
(e)	Issue of new shares	Current ratio will improve	As cash will increase, current assets will increase and current ratio will increase.

$\text{Eq.} = \frac{\text{Assets}}{\text{Sh. Eq.}} = \frac{120}{40} = 3$

$\text{ROE} = \text{NP Ratio} \times \text{Assets} \times \text{Eq. Mul.}$
 $= 0.12 \times 2.5 \times 3$
 $= 0.9$

Question – 3

A Limited Company's books reveal following information: Net Income

Net Income	→ ₹ 3,60,000
Shareholder's Equity	→ ₹ 4,00,000
Assets Turnover	→ 2.5 times
Net profit margin	→ 12%

$\text{Sales} = \frac{3.60 \text{ l}}{12\%} = 30 \text{ l}$
 $\frac{\text{Sales}}{\text{Assets}} = 2.5 \Rightarrow \text{Assets} = \frac{30 \text{ l}}{2.5} = 12 \text{ l}$

You are required to calculate ROE (Return on Equity) of the company based on the 'DuPont Model'.

Solution

Net Profit Margin = Net Income ÷ Revenue

$0.12 = 3,60,000 \div \text{Revenue}$

Revenue = ₹ 30,00,000

Assets turnover = Revenue ÷ Assets

$2.5 = 30,00,000 \div \text{Assets}$

Assets = ₹ 12,00,000

$$\text{Equity Multiplier} = \text{Assets} \div \text{Shareholder's Equity} = 12,00,000 \div 4,00,000 = 3$$

$$\text{Return on Equity} = \text{Net Profit Margin} \times \text{Assets Turnover} \times \text{Equity Multiplier}$$

$$= 0.12 \times 2.5 \times 3 = 0.9 = 9\%$$

$$\frac{A}{E.A.} = \frac{1.5}{1} \Rightarrow E.A. = \frac{450}{1.5} = 300$$

$$54\% = \frac{PAT}{Sh.}$$

$$PBT = \frac{16200}{1-0.25}$$

$$EBIT = 21600 + 56250$$

Question – 4

$$PAT = 300 \times 54\% = 16200$$

$$= 21600$$

$$= 2722500$$

Theme Ltd provides you the following information:

12.5% Debt	→	✓ ₹45,00,000
Debt to Equity ratio	→	✓ 1.5:1
Return on Shareholder's fund	→	✓ 54%
Operating Ratio	→	85%
Ratio of operating expenses to Cost of Goods sold	→	2:6
Tax rate	→	✓ 25%
Fixed Assets	→	✓ ₹39,00,000
Current Ratio	→	✓ 1.8

You are required to calculate:

(i) Interest Coverage Ratio

(ii) Gross Profit Ratio

(iii) Current Assets

$$\frac{EBIT}{Int.} = \frac{2722500}{(450 \times 12.5\%)} = \frac{2722500}{56250} = 4.84$$

$$\frac{GP}{Sales} = \frac{6579375}{181.52} = 36.25\%$$

$$\rightarrow 450 \times 1.8 = 810$$

$$C.R. = \frac{CA}{CL} \Rightarrow 1.8 = \frac{CA}{CL}$$

$$CA = (1.8)(CL)$$

$$Op. Pbt. = 15\% \rightarrow Sales = \frac{2722500}{15\%} = 181.52$$

$$Op. Cost = 181.52 \times 85\% = 15427500$$

$$Op. Exp. (2:6) \text{ CoGS}$$

$$3856875 \quad 11570625$$

$$GP = 181.52 - 11570625 = 6579375$$

Solution**Working Notes:**

$$\text{Debt} = ₹45,00,000$$

$$\text{Interest} = ₹45,00,000 \times 12.5\% = ₹5,62,500$$

$$\text{Debt to Equity} = 1.5:1 = \frac{\text{Total Debt}}{\text{Shareholders' Equity}}$$

$$\text{Equity} = ₹30,00,000$$

$$\text{Return of Shareholder's funds} = 54\% = \frac{\text{Net Profit after taxes}}{\text{Equity share holders fund}} \times 100$$

$$\text{Profit after tax (PAT)} = 54\% \times \text{Equity} = ₹16,20,000$$

$$\text{Profit before tax (PBT)}(1-25\%) = \text{Profit after tax}$$

$$= ₹16,20,000 / 75\% = ₹21,60,000$$

$$\text{Earning before interest and tax (EBIT)} = \text{PBT} + \text{Interest}$$

$$= ₹21,60,000 + ₹5,62,500 = ₹27,22,500$$

(i) Interest Coverage Ratio

$$= \text{EBIT} / \text{Interest}$$

$$= ₹27,22,500 / ₹5,62,500 = 4.84 \text{ Times}$$

(ii) Operating Profit Ratio

$$= 1 - \text{Operating Ratio}$$

$$= 1 - 0.85 = 0.15 \text{ or } 15\%$$

$$0.15 = \frac{\text{Operating Profit}}{\text{Sales}} \times 100$$

$$\text{Sales} = \text{EBIT or Operating Profit} / 0.15$$

$$= ₹27,22,500 / 0.15 = ₹1,81,50,000$$

$$\text{Operating ratio} = \frac{\text{Operating expenses}}{\text{Cost of Goods Sold (COGS)}} = 2:6 = 1:3$$

Operating expenses	= 1/3COGS
Operating cost	= Sales – Operating profit
	= ₹1,81,50,000 – ₹27,22,500 = ₹1,54,27,500
₹1,54,27,500	= COGS + Operating expenses
₹1,54,27,500	= COGS + 1/3COGS
COGS	= ₹1,15,70,625
Gross profit	= Sales – COGS
	= 1,81,50,000 – 1,15,70,625 = ₹65,79,375
Gross Profit ratio	= $\frac{\text{Gross Profit}}{\text{Sales}} \times 100$
	= 65,79,375/1,81,50,000 = 0.3625 or 36.25%

Gross profit and sales can be calculated in alternative way also. However, there will be no change in GP ratio i.e. 36.25%

(iii) Current Ratio	= $\frac{\text{Current Assets}}{\text{Current Liabilities}} = 1.8$
Current Assets	= 1.8 Current Liabilities
Total of Balance sheet liability	= Equity + Debt + Current Liabilities
	= 30,00,000 + 45,00,000 + CL (2)
Total Balance sheet asset	= Fixed Assets + Current Assets
	= 39 lakhs + CA = 39 + 1.8CL (3)

Equating 2 and 3,

75,00,000 + CL	= 39,00,000 + 1.8CL
0.8CL	= 36,00,000
CL	= ₹45,00,000
Current Assets	= 1.8 CL = 1.8 × 45 lakhs = ₹81,00,000

Question – 5

With the following ratios and further information given below prepare a Trading Account, Profit and Loss Account and Balance Sheet of ABC Company.

Fixed Assets	→ ₹ 40,00,000
Closing stock	→ ₹ 4,00,000 ✓
Stock turnover ratio	10
Gross profit ratio	25 percent → $GP = 32l \times 25\% = 8l$
Net profit ratio	20 percent → $Sales = 6.40l \div 20\% = 32l$
Net profit to capital	1/5 → $NP = Cap \times 1/5 = 32l \times 1/5 = 6.40l$
Capital to total liabilities	1/2 → $\frac{Cap}{TL} = \frac{1}{2} \Rightarrow TL = Cap \times 2 = 32l \times 2 = 64l$
Fixed assets to capital	→ 5/4 → $\frac{40l}{Cap} = \frac{5}{4} \Rightarrow Cap = 32l$
Fixed assets / Total current assets	5/7 → $\frac{40l}{CA} = \frac{5}{7} \Rightarrow CA = 56l$

$$\begin{aligned}
 STP &= \frac{COGS}{\text{Avg. St.}} \\
 10 &= \frac{(32l - 8l)}{\text{Avg. St.}} \\
 \text{Avg. St.} &= 2.40l
 \end{aligned}
 \quad \left| \quad
 \begin{aligned}
 \text{Avg. St.} &= \frac{Op. + Cl}{2} \\
 2.40l &= \frac{Op. + 4l}{2} \\
 Op. &= 0.80l
 \end{aligned}$$

Solution**Trading and P&L Account**

To Opening Stock (w.n. – 9)	80,000	By Sales (w.n. – 5)	32,00,000
To Purchases (bal. fig.)	27,20,000	By Closing Stock (given)	4,00,000
To Gross Profit (w.n. – 6)	8,00,000		
	36,00,000		36,00,000
To Expenses (bal. fig.)	1,60,000	By Gross Profit b/d	8,00,000
To Net Profit (w.n. – 4)	6,40,000		
	8,00,000		8,00,000

Balance Sheet

Capital (w.n. – 1)	32,00,000	Fixed Assets (given)	40,00,000
Other Liabilities (w.n. – 3)	64,00,000	Current Assets:	
		Stock (given)	4,00,000
		Other CA (w.n. – 10)	52,00,000
	96,00,000		96,00,000

Working Notes:

- $\frac{\text{Fixed Assets}}{\text{Capital}} = \frac{40,00,000}{\text{Capital}} = \frac{5}{4}$. Thus, Capital = ₹ 32,00,000
- $\frac{\text{Fixed Assets}}{\text{Total Current Assets}} = \frac{40,00,000}{\text{Total Current Assets}} = \frac{5}{7}$. Thus, Total Current Assets = ₹ 56,00,000
- $\frac{\text{Capital}}{\text{Other liabilities}} = \frac{32,00,000}{\text{Other liabilities}} = \frac{1}{2}$. Thus, Other Liabilities = ₹ 64,00,000
Total liabilities is taken/assumed as “External Liabilities”, i.e. excluding capital.
- $\frac{\text{Net Profit}}{\text{Capital}} = \frac{\text{Net Profit}}{32,00,000} = \frac{1}{5}$. Thus, Net Profit = ₹ 6,40,000
- Net Profit ratio = $\frac{\text{Net Profit}}{\text{Sales}} = \frac{6,40,000}{\text{Sales}} = 20\%$. Thus, Sales = ₹ 32,00,000
- Gross Profit Ratio = 25%. Thus, Gross profit = $32,00,000 \times 25\% = ₹ 8,00,000$
- Cost of goods sold = Sales – Gross profit = $32,00,000 - 8,00,000 = ₹ 24,00,000$
- Stock Turnover ratio = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{24,00,000}{\text{Average Stock}} = 10$.
Thus, Average Stock = ₹ 2,40,000
- Average Stock = $\frac{\text{Opening stock} + \text{closing stock}}{2} = \frac{\text{Opening stock} + 4,00,000}{2} = 2,40,000$.
Thus, Op. Stock = ₹ 80,000
- Other Current Assets = Total Current Assets – Stock = $56,00,000 - 4,00,000 = ₹ 60,00,000$

Question – 6

Following is the abridged balance Sheet of the SK Ltd. as at 31st March, 2020:

Balance Sheet

Liabilities	₹	Assets	₹
Paid up share capital	✓ 5,00,000	Free hold property	4,00,000 ✓
Profit & Loss A/c	85,000	Plant & Machinery	2,50,000 ✓
Current Liabilities	2,00,000	Less: Acc. Depreciation	75,000
		Stock	1,05,000
		Debtors	1,00,000
		Bank	5,000
	7,85,000		7,85,000

From the following information you are required to prepare Profit & Loss A/c (2020-21) and Balance Sheet as at 31st March, 2021:

- (a) The composition of the total of 'liabilities' side to the company's Balance Sheet as at 31st March, 2021 (the paid-up capital remaining the same as at 31st March, 2020) was:

Share Capital
Profit & Loss A/c
10% Debentures
Creditors

→ 50%
→ 15%
→ 10%
→ 25%

$$\frac{1}{1} = \frac{CA - Stock}{CL}$$

$$2.50L = 4L - Stock$$

$$Stock = 1.50L$$

$$Depr. = \frac{4}{5} \times (4L - 1.50L)$$

$$= 2L$$

The debentures were issued on 1st April, 2020. Interest is to be paid on 30th September, 2020 and 31st March, 2021. $Int. = 10\% \times 1L = 10000$

- (b) During the year ended on 31st March, 2021, additional Plant and Machinery had been bought and a further ₹ 25,000 depreciation written off. Freehold property remained unchanged. The total fixed assets then constituted 60% of total fixed and current assets. $\rightarrow FA = 10L \times 60\% = 6L$ $CA = 4L$
- (c) The current ratio was 1.6:1. The quick ratio was 1:1 $Sales = \frac{2L}{2} \times 12 = 12L$
- (d) The debtors (four-fifth of the quick assets) to sales ratio revealed a credit period of two months.
- (e) Gross Profit was at the rate of 15% of selling price and return on net worth as at 31st March, 2021 was 10%. Ignore taxation.

$$10\% = \frac{PAT}{Sh. fund} \Rightarrow PAT = 10\% \times 6.5L$$

$$= 65000$$

Solution**Working Notes:**

(1)	Particulars	%	Amount
	Share capital	50%	5,00,000 (Because it remains same for next year)
	P&L Account	15%	1,50,000 (10,00,000 × 15%)
	10% Debentures	10%	1,00,000 (10,00,000 × 10%)
	Creditors	25%	2,50,000 (10,00,000 × 25%)
			10,00,000 (5,00,000 ÷ 50%)

$$(2) \text{ Interest on debenture} = 1,00,000 \times 10\% = ₹ 10,000$$

$$(3) \text{ Total assets} = \text{Total liabilities} = ₹ 10,00,000$$

$$\text{Fixed assets} = 60\% \times \text{Total assets} = 60\% \times 10,00,000 = ₹ 6,00,000$$

$$\text{Current assets} = 10,00,000 - 6,00,000 = ₹ 4,00,000$$

$$(4) \text{ Fixed assets} = \text{Freehold property} + \text{Plant \& Machinery}$$

$$6,00,000 = 4,00,000 + \text{Plant \& Machinery}$$

$$\text{Plant \& Machinery} = ₹ 2,00,000$$

$$(5) \text{ Current ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$1.60 = \frac{4,00,000}{\text{Current Liabilities}}$$

$$\text{Current Liabilities} = ₹ 2,50,000$$

$$(6) \text{ Quick ratio} = \frac{\text{Liquid assets}}{\text{Liquid liabilities}}$$

$$1 = \frac{4,00,000 - \text{Stock}}{2,50,000}$$

$$\text{Stock} = ₹ 1,50,000$$

$$(7) \text{ Debtors} = \frac{4}{5} \times \text{Quick Assets} = \frac{4}{5} \times (4,00,000 - 1,50,000) = ₹ 2,00,000$$

$$(8) \text{ Other current assets} = 4,00,000 - 1,50,000 - 2,00,000 = ₹ 50,000$$

$$(9) \text{ Credit period} = \frac{\text{Average Debtors}}{\text{Credit Sales}} \times 12$$

$$2 = \frac{2,00,000}{\text{Credit sales}} \times 12$$

$$\text{Credit sales} = ₹ 12,00,000$$

$$(10) \text{ Gross Profit} = 15\% \times \text{Sales} = 15\% \times 12,00,000 = ₹ 1,80,000$$

$$(11) \text{ Return on net worth} = \frac{\text{Profit}}{\text{Shareholder fund}}$$

$$\frac{10}{100} = \frac{\text{Profit}}{5,00,000 + 1,50,000}$$

$$\text{Profit} = ₹ 65,000$$

Trading and Profit and Loss Account

Particulars	₹	Particulars	₹
To Opening Stock	1,05,000	By Sales	12,00,000
To Purchases & Other exp. (Bal. fig.)	10,65,000	By Closing Stock	1,50,000
To Gross Profit c/d	1,80,00		
	13,50,000		13,50,000
To Depreciation	25,000	By Gross Profit b/d (Bal. fig.)	1,80,000
To Interest on Debentures	10,000		
To Other exp. (Bal. fig.)	80,000		
To Net profit for the year c/d	65,000		
	1,80,000		1,80,000

Balance Sheet

Liabilities	₹	Assets	₹
Share Capital	5,00,000	<u>Fixed assets:</u>	
Profit & loss A/c	1,50,000	Freehold property	4,00,000
10% Debentures	1,00,000	Plant & Machinery	<u>2,00,000</u>
Creditors	2,50,000	<u>Current assets:</u>	
		Stock	1,50,000
		Debtors	2,00,000
		Other CA	<u>50,000</u>
	10,00,000		4,00,000
			10,00,000

$$10l = 3(CL) - CL$$

$$CL = 5l \therefore CA = 15l$$

$$\frac{1}{1} = \frac{15l - 5l}{5l}$$

$$Cl. St. = 10l$$

Question - 7

From the following information and ratios, prepare the balance sheet as at 31st march, 2023 and Income statement for the year ended on that date for M/s SK Ltd.

Average stock ✓	→ ₹ 10 lakhs
Current ratio	3:1 → CA = 3(CL)
Acid test ratio	1:1
PBIT to PBT	2.2:1
Average collection period (assume 360 days in a year) → $\frac{360}{\frac{Sales}{PBT}} = 30 \text{ days}$	30 days
Stock turnover ratio (use sales as turnover)	5 times = $\frac{Sales}{\text{Stock}} \Rightarrow Sales = 5 \times \text{Stock}$
Fixed assets turnover ratio → $FA = \frac{Sales}{\frac{Sales}{0.8}} = \frac{50l}{0.8} = 62.5l$	→ 0.8 times
Working capital	→ ₹ 10 lakhs = CA - CL
Net profit ratio	10% → $50l \times 10\% = 5l$
Gross profit ratio	40% → $50l \times 40\% = 20l$
Operating expenses (excluding interest)	→ ₹ 9 lakhs
Long term loan interest	12%
Tax	Nil

Solution

- Current Ratio = 3:1
 Current Assets (CA)/Current Liability (CL) = 3:1
 $CA = 3CL$
 $WC = 10,00,000$
 $CA - CL = 10,00,000$
 $3CL - CL = 10,00,000$
 $2CL = 10,00,000$
 $CL = \frac{10,00,000}{2}$
 $CL = ₹ 5,00,000$
 $CA = 3 \times 5,00,000$
 $CA = ₹ 15,00,000$

2. Acid Test Ratio = $CA - \text{Stock} / CL = 1:1$
 $= \frac{15,00,000 - \text{Stock}}{5,00,000} = 1$
 $15,00,000 - \text{stock} = 5,00,000$
 $\text{Stock} = ₹10,00,000$
3. Stock Turnover ratio (on sales) = 5
 $\frac{\text{Sales}}{\text{Avg stock}} = 5$
 $\frac{\text{Sales}}{10,00,000} = 5$
 $\text{Sales} = ₹50,00,000$
4. Gross Profit = $50,00,000 \times 40\% = ₹20,00,000$
 Net profit (PBT) = $50,00,000 \times 10\% = ₹5,00,000$
5. PBIT/PBT = 2.2
 $\text{PBIT} = 2.2 \times 5,00,000$
 $\text{PBIT} = 11,00,000$
 $\text{Interest} = 11,00,000 - 5,00,000 = ₹6,00,000$
 $\text{Long term loan} = \frac{6,00,000}{0.12} = ₹50,00,000$
6. Average collection period = 30 days
 $\text{Receivables} = \frac{30}{360} \times 50,00,000 = 4,16,667$
7. Fixed Assets Turnover Ratio = 0.8
 $50,00,000 / \text{Fixed Assets} = 0.8$
 $\text{Fixed Assets} = ₹62,50,000$

Income Statement

	(₹)
Sales	50,00,000
Less: cost of goods sold	30,00,000
Gross profit	20,00,000
Less: operating expenses	9,00,000
Less: Interest	6,00,000
Net Profit	5,00,000

**Balance sheet**

Liabilities	₹	Assets	₹
Equity Share Capital	22,50,000	Fixed assets	62,50,000
Long term debt	50,00,000	Current Assets:	
Current liability	5,00,000	Stock 10,00,000	
		Cash 4,16,667	
		Other 83,333	15,00,000
Total	77,50,000	Total	77,50,000

$$0.9 = \frac{CA - 97778}{112} \Rightarrow CA = 196778$$

Question – 8

Following are the data in respect of ABC Industries for the year ended 31st March, 2021:

Debt to Total assets ratio	:	0.40 $\rightarrow Debt = 502 \times 0.4 = 202$
Long-term debts to equity ratio	:	30% $\rightarrow 30\% \times (202 + 102) = 92$
Gross profit margin on sales	:	20% $\rightarrow 642 = 80\% \rightarrow Sales = \frac{642}{80\%} = 802$
Accounts receivables period	:	36 days $\rightarrow 802 \times \frac{36}{360} = 82$
Quick ratio	:	0.9
Inventory holding period	:	55 days $\rightarrow 642 \times \frac{55}{360} = 97778$
Cost of goods sold	:	$\rightarrow ₹ 64,00,000$

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets (B/F)	30,32,222
Reserve & surplus $502 - 202 - 202 = 102$	102	Inventories	97778
Long-term debts	92	Accounts receivable	8,00,000
Accounts payable $202 - 92 = 112$	112	Cash $196778 - 82 - 97778 = 190000$	190000
Total	50,00,000	Total	50,00,000

Required:

Complete the balance sheet of ABC Industries as on 31st March, 2021. All calculations should be in nearest rupee. Assume 360 days in a year.

Solution

Balance Sheet of ABC Industries as on 31st March, 2021

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets	30,32,222
Reserve & surplus	10,00,000	Inventories	9,77,778
Long-term debts	9,00,000	Accounts receivable	8,00,000
Accounts payable	11,00,000	Cash	1,90,000
Total	50,00,000	Total	50,00,000

Note:

Working Notes:

- (1) Total liabilities = Total assets = ₹ 50,00,000

$$\frac{Debt}{Total Assets} = 0.40$$

$$\frac{Debt}{50,00,000} = 0.40$$

$$Debt = ₹ 20,00,000$$

- (2) Reserve & Surplus = Total liabilities – Equity capital – Debt
 $= 50,00,000 - 20,00,000 - 20,00,000 = ₹ 10,00,000$

- (3) $\frac{Long term debt}{Equity shareholder fund} = 30\%$

$$\frac{Long term debt}{(20,00,000 + 10,00,000)} = 30\%$$

$$Long term debt = ₹ 9,00,000$$

- (4) Accounts payable = total debt - long term debt = 20,00,000 – 9,00,000 = ₹ 11,00,000

- (5) COGS ratio = 100 – GP Ratio = 100 – 20% = 80% of sales

- (6) Sales = $\frac{\text{Cost of goods sold}}{\text{COGS Ratio}} = \frac{64,00,000}{80\%} = ₹ 80,00,000$
- (7) Closing inventory = $\frac{\text{Cost of goods sold}}{\text{Inventory days}} \times 360 = \frac{64,00,000}{55} \times 360 = ₹ 9,77,778$
- (8) Account receivables = $\frac{\text{Credit sales}}{\text{Account receivable period}} \times 360 = \frac{80,00,000}{36} \times 360 = ₹ 8,00,000$
- (9) Quick ratio = $\frac{\text{Quick assets}}{\text{Current liabilities}}$
 $0.90 = \frac{\text{Cash} + \text{Debtors}}{11,00,000}$
 Cash + 8,00,000 = 9,90,000
 Cash = ₹ 1,90,000
- (10) Fixed assets = Total assets – current assets
 $= 50,00,000 - (9,77,778 + 8,00,000 + 1,90,000) = ₹ 30,32,222$

$$CA = \frac{3}{6} + \frac{3}{6} + \frac{58}{3} = \frac{3}{3} + \frac{58}{3}$$

$$\begin{aligned} C + CP &= \text{Sales} \\ 100 + 20 &= 120 \\ &\quad \quad \quad \frac{1}{6} \end{aligned}$$

Question – 9

From the following information, you are required to prepare a summarized balance sheet for Rudra Ltd. for the year 31st March, 2023:

Debt Equity Ratio

Current ratio

Acid test ratio

Fixed assets turnover (on the basis of sales)

Stock turnover (on the basis of sales)

Cash in hand

Stock to debtors

Sales to net worth

Capital to reserves

Gross profit

COGS to creditor

$$1:1 \rightarrow \text{Debt} = \frac{3}{4}$$

$$3:1 \rightarrow CL = \frac{CA}{3} = \left(\frac{3}{2} + 58\right) \left(\frac{1}{3}\right) = \frac{3}{9} + \frac{58}{3}$$

$$8:3$$

$$4 \rightarrow FA = \frac{3}{4}$$

$$6 \rightarrow \text{Stock} = \frac{3}{6} \checkmark$$

$$₹ 5,00,000$$

$$1:1 \rightarrow \text{Debt} = \frac{3}{6} \checkmark$$

$$4 \rightarrow NW = \frac{3}{4}$$

$$1:2$$

$$20\% \text{ of cost} \rightarrow CP = \frac{3}{6} \quad \text{Cost} = \frac{58}{6}$$

$$10:1 \rightarrow \frac{\text{Cost}}{\text{Cred.}} = \frac{10}{1} \Rightarrow \text{Cred.} = \frac{58}{60} = \frac{3}{12}$$

Interest for entire year is yet to be paid on long term loan @10%.

Solution**Balance Sheet of Rudra Ltd.**

Liabilities	₹	Assets	₹
Capital	10,00,000	Fixed Assets	30,00,000
Reserves	20,00,000	Current Assets:	
Long term loan @10%	30,00,000	Stock in trade	20,00,000
Current Liabilities:		Debtors	20,00,000
Creditors	10,00,000	Cash	5,00,000
Other STCL	2,00,000		
Outstanding Interest	3,00,000		
	75,00,000		75,00,000

Working Notes:

Let sales be ₹x

Balance Sheet of Rudra Ltd.

Liabilities	₹	Assets	₹
Capital		Fixed Assets	x/4
Reserves		Current Assets:	
Net worth	x/4	Stock in trade	x/6
Long term loan @10%	x/4	Debtors	x/6
Current Liabilities:		Cash	5,00,000
Creditors	x/12		
Other STCL			
Outstanding Interest	3,00,000		
Total Current liabilities	(x/9 + 5,00,000/3)		

$$1. \quad \text{Fixed Asset Turnover} = 4 = \frac{x}{\text{Fixed Assets}}$$

$$\text{Fixed Assets} = \frac{x}{4}$$

$$2. \quad \text{Stock Turnover} = 6 = \frac{x}{\text{stock}}$$

$$\text{Stock} = \frac{x}{6}$$

$$3. \quad \text{Sales to net worth} = 4 = \frac{x}{\text{net worth}}$$

$$\text{Net worth} = \frac{x}{4}$$

$$4. \quad \text{Debt: Equity} = 1:1$$

$$\frac{\text{Long Term Loan}}{\text{Net worth}} = \frac{1}{1}$$

$$\text{Long term loan} = \text{Net worth} = \frac{x}{4}$$

$$5. \quad \text{Gross Profit to Cost} = 20\%$$

$$\frac{\text{GP}}{\text{Sales-GP}} = 20\%$$

$$\frac{\text{GP}}{x-\text{GP}} = 20\%$$

$$\text{GP} = 0.2x - 0.2\text{GP}$$

$$1.2\text{GP} = 0.2x$$

$$\text{GP} = \frac{0.2x}{1.2}$$

$$\text{GP} = x/6$$

$$\text{Cost of Goods Sold} = x - x/6 = 5/6x$$

$$6. \quad \text{COGS to creditors} = 10:1$$

$$\frac{\text{COGS}}{\text{Creditors}} = \frac{10}{1}$$

$$\frac{\frac{5}{6}x}{\text{Creditors}} = \frac{10}{1}$$

$$\text{Creditors} = \frac{5x}{60} = \frac{x}{12}$$

$$7. \quad \frac{\text{Stock}}{\text{Debtor}} = 1$$

$$\text{Debtor} = \text{Stock} = \frac{x}{6}$$

8. Current Ratio = 3: 1

$$\frac{\text{Stock} + \text{Debtors} + \text{Cash}}{\text{Current Liabilities}} = \frac{3}{1}$$

$$\frac{\frac{x}{6} + \frac{x}{6} + 5,00,000}{\text{Current Liabilities}} = 3$$

$$\frac{\frac{x}{3} + 5,00,000}{3} = CL$$

$$CL = \frac{x}{9} + \frac{5,00,000}{3}$$

9. CA = 3CL

$$= 3 \left(\frac{x}{9} + \frac{5,00,000}{3} \right)$$

$$CA = \frac{x}{3} + 5,00,000$$

10. Net worth + Long Term Loan + Current Liability = Fixed Asset + Current Assets

$$\frac{x}{4} + \frac{x}{4} + \frac{x}{9} + \frac{5,00,000}{3} = \frac{x}{4} + \frac{x}{3} + 5,00,000$$

$$\frac{x}{4} + \frac{x}{9} - \frac{x}{3} = 5,00,000 - \frac{5,00,000}{3}$$

$$\frac{9x + 4x - 12x}{36} = \frac{15,00,000 - 5,00,000}{3}$$

$$\frac{x}{36} = \frac{10,00,000}{3}$$

$$x = ₹1,20,00,000$$

11. Now, from above calculations, we get,

$$\text{Fixed Asset} = \frac{x}{4} = \frac{₹1,20,00,000}{4} = ₹30,00,000$$

$$\text{Stock} = \frac{x}{6} = \frac{₹1,20,00,000}{6} = ₹20,00,000$$

$$\text{Debtor} = \frac{x}{6} = \frac{₹1,20,00,000}{6} = ₹20,00,000$$

$$\text{Net Worth} = \frac{x}{4} = ₹30,00,000$$

Now, Capital to Reserve is 1: 2

$$\text{Capital} = ₹10,00,000$$

$$\text{and, Reserve} = ₹20,00,000$$

$$\text{Long Term Loan} = \frac{x}{4} = 30,00,000$$

$$\text{Outstanding Interest} = 30,00,000 \times 10\% = 3,00,000$$

$$\text{Creditors} = \frac{x}{12} = \frac{₹1,20,00,000}{12} = ₹10,00,000$$

$$\text{Current Liabilities} = \text{Creditors} + \text{Other STCL} + \text{Outstanding Interest}$$

$$\frac{x}{9} + \frac{5,00,000}{3} = ₹10,00,000 + \text{Other STCL} + ₹3,00,000$$

$$\frac{₹1,20,00,000}{9} + \frac{5,00,000}{3} = ₹13,00,000 + \text{Other STCL}$$

$$₹15,00,000 = \text{Other STCL} + ₹13,00,000$$

$$\text{Other STCL} = ₹2,00,000$$